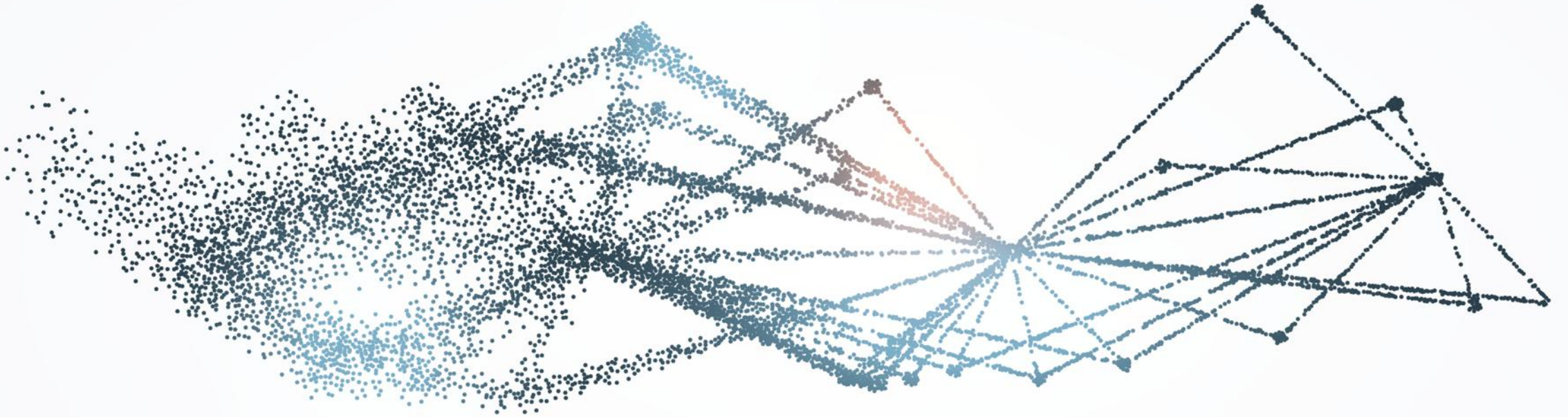




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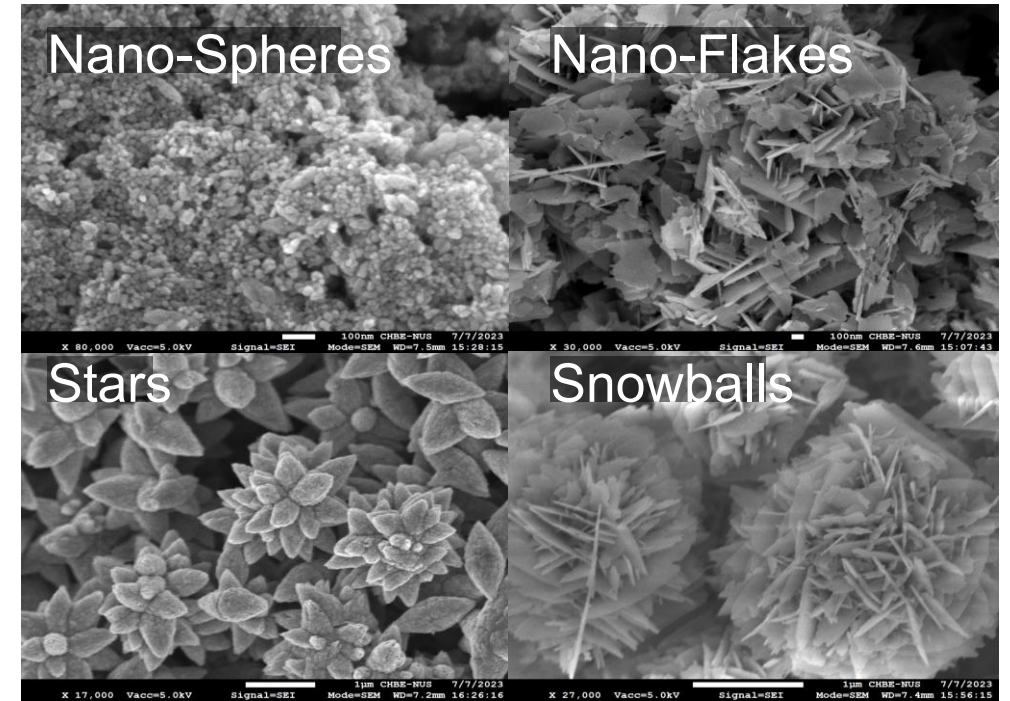
Materials List

Zinc oxide, CAS # 1314-13-2

Zinc oxide (ZnO) is a versatile inorganic compound known for its wide range of applications. It appears as a white powder and is used in various industries, including cosmetics, where it provides UV protection and skin soothing properties; electronics, for its semiconducting and piezoelectric characteristics; and as an additive in products like rubber, paints, and ceramics. ZnO also exhibits antimicrobial properties, making it valuable in medical and personal care products. Its unique combination of properties makes it essential in both industrial and consumer applications.

Property	Spheres	Flakes	Stars	Snowballs
Formula	ZnO	ZnO	ZnO	ZnO
Crystal phase	Zincite	Zincite	Zincite	Zincite
Particle size, nm	30	150	1500	1500
Purity, %	98+	97+	99+	97+
Heavy metals content, ppm	<5	<10	<10	<5
Application	Cosmetics	Paints	-	Cosmetics
Function	UV-block	Antimicrobial, white pigment	-	Antiperspirant

ZnO morphologies



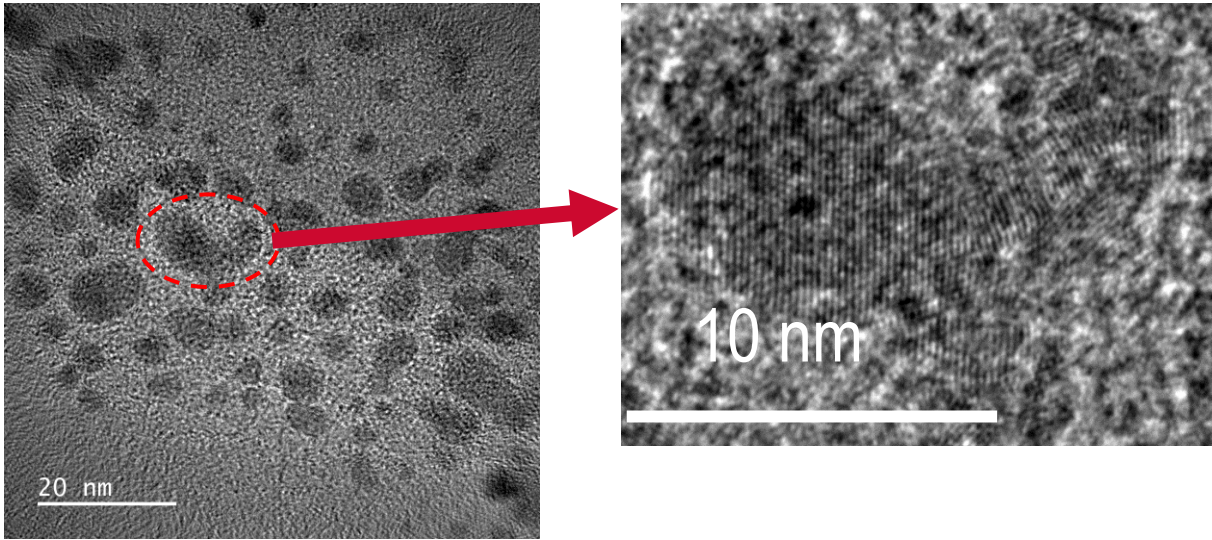
Reference: <https://doi.org/10.1016/j.cej.2021.131345>

Zirconium (IV) Dioxide, CAS # 1314-23-4

Zirconium(IV) dioxide, also known as zirconia (ZrO_2), is a white crystalline oxide of zirconium. It is highly valued for its high melting point, thermal stability, and strong resistance to corrosion. Zirconia is commonly used in ceramics, where it provides toughness and durability, and in dental applications for making crowns and bridges due to its biocompatibility and natural tooth-like appearance. It is also used as a refractory material and in oxygen sensors and fuel cells due to its ability to conduct oxygen ions.

Property	Spheres
Formula	ZrO_2
Crystal phase	Cubic
Particle size, nm	<5
Purity, %	98+
Heavy metals content, ppm	<5
Application	Optics
Function	High-refractive index coatings

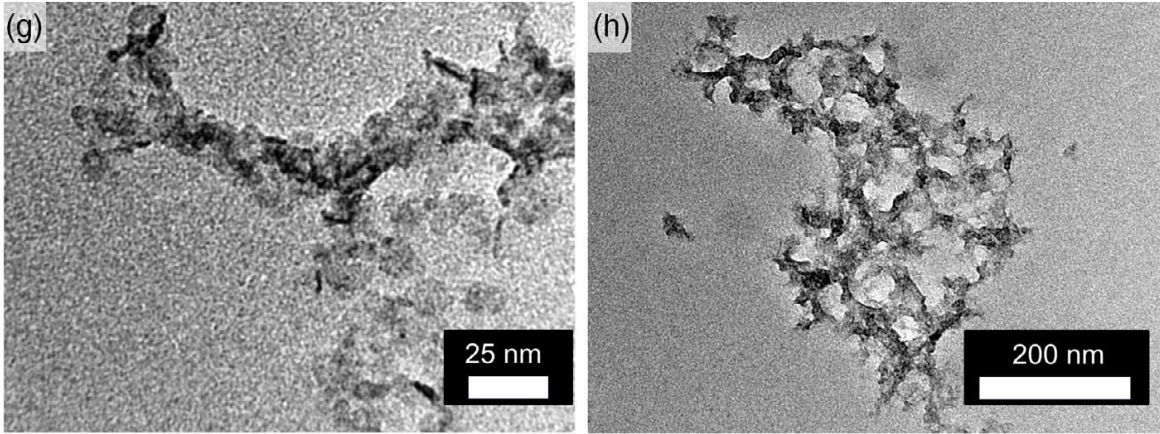
HR-TEM of zirconia



Nickel-Cobalt Hydroxide, CAS # 61179-08-6

Nickel-Cobalt Hydroxide is a mixed-metal hydroxide compound known for its role in energy storage technologies. It is primarily used as a precursor material in the production of battery electrodes, particularly in lithium-ion and nickel-metal hydride batteries. The material provides high specific capacity and improved cycling stability, making it a crucial component for enhancing the performance of rechargeable batteries.

Property	Flakes
Formula	NiCo(OH) ₄
Crystal phase	α-NiCo-hydroxide
Particle size, nm	12
Purity, %	98+
Application	Supercapacitors
Function	Charge-storage
Specific capacity, F g ⁻¹	4700
Stability, # of cycles	> 20000



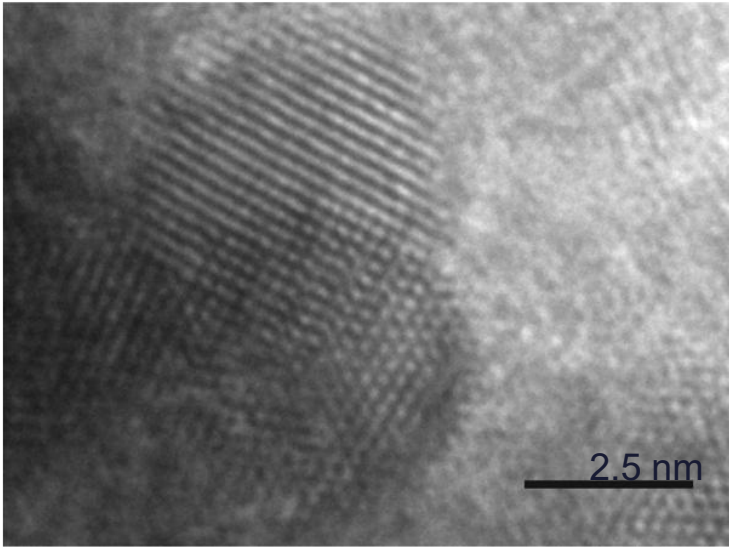
HR-TEM of NiCo-hydroxide

Reference: <https://doi.org/10.1002/ente.202200633>

Hydrotalcite, CAS # 11097-59-9

Hydrotalcite is a layered double hydroxide mineral composed primarily of magnesium and aluminum hydroxides. It is known for its unique structure, which allows it to act as an anion exchanger. Hydrotalcite is widely used in industrial applications, including as a stabilizer in polymers like PVC, as an antacid in pharmaceuticals, and as a precursor for catalysts in chemical reactions. Its ability to absorb and release various anions makes it valuable in environmental remediation and as an additive to improve the thermal stability of materials.

Property	Flakes
Formula	MgAl(OH) _x
Crystal phase	Monoclinic
Particle size, nm	5-28
Application	Additive, adsorbent
Function	Fire retardant additive, gas adsorbents



HR-TEM of Hydrotalcite

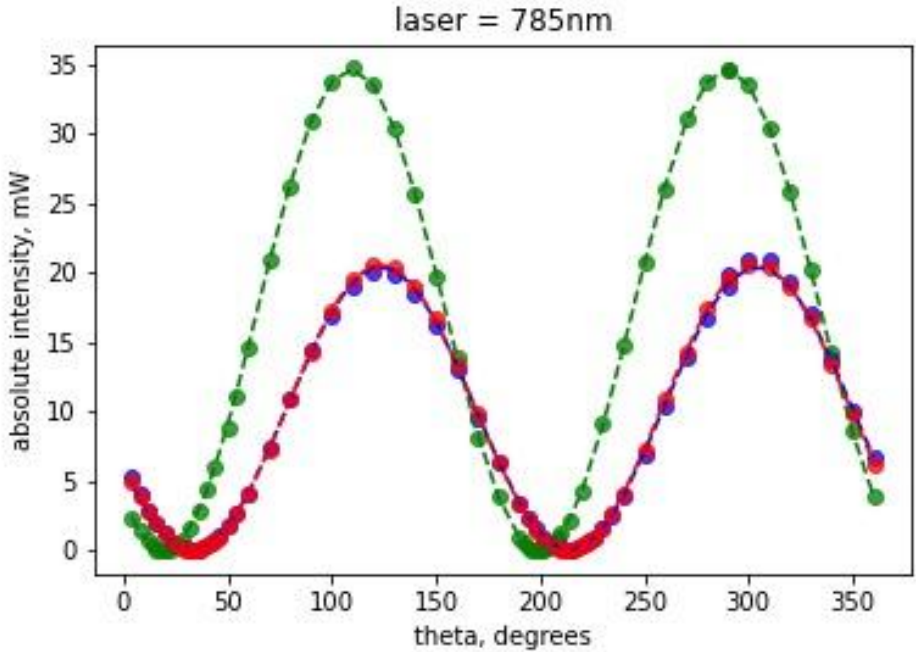
Reference: <https://doi.org/10.1038/s41467-018-07395-4>

Cobalt-Iron oxide, CAS # 12052-28-7

Cobalt-Iron oxide, commonly referred to as cobalt ferrite (CoFe_2O_4), is a magnetic material known for its high coercivity, moderate magnetization, and chemical stability. This mixed-metal oxide is used in various applications, including magnetic storage devices, spintronics, and as a contrast agent in magnetic resonance imaging (MRI). Its unique combination of magnetic properties makes it valuable in the production of permanent magnets, magneto-optical devices, and as a catalyst in chemical processes

Magneto-optical sensing of composite film with CoFe_2O_4 . Green – initial polarization, blue – polarization of the film, and red – polarization after a small magnetic field is applied by a magnet. The shift between the red and blue curves ($\sim 0.1 - 0.2$ degrees) matches previous studies. This corresponds to a Verdet constant, $V = 10^5 \text{ rad}/(\text{T}\cdot\text{m})$.

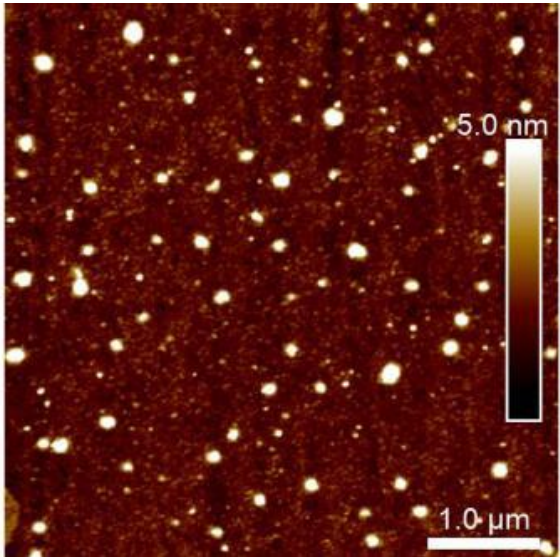
Property	Flakes
Formula	CoFe_2O_4
Crystal phase	$\alpha\text{-Fe}_2\text{O}_3$ and Co_3O_4
Particle size, nm	12-25
Purity, %	98+
Application	Nanomagnets
Function	Magnetic sensors



Curcumin, CAS # 458-37-7

Curcumin is a natural polyphenolic compound found in the spice turmeric (*Curcuma longa*). It is well-known for its bright yellow color and is widely used as a food coloring and flavoring agent. Beyond its culinary uses, curcumin has garnered significant attention for its potential health benefits due to its anti-inflammatory, antioxidant, and anticancer properties. It is studied for its therapeutic potential in treating various conditions, including arthritis, cardiovascular diseases, and neurodegenerative disorders. However, curcumin has low bioavailability, which has led to the development of various formulations to enhance its absorption and efficacy in the body.

Property	Spheres
Formula	C ₂₁ H ₂₀ O ₆
Particle size, nm	230
Purity, %	98+
Application	Nutrient
Function	Increased bioavailability



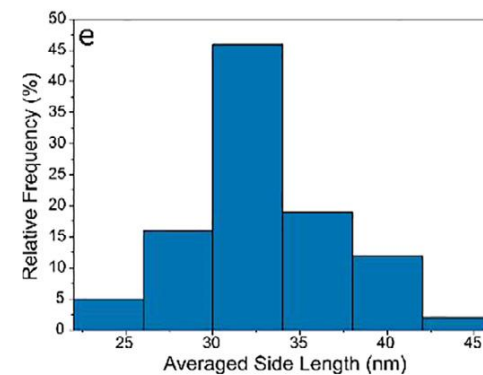
AFM image of curcumin nanoparticles deposited on freshly cleaved mica slides

Reference: Master's thesis Lechtenberg et al (University of Cambridge)

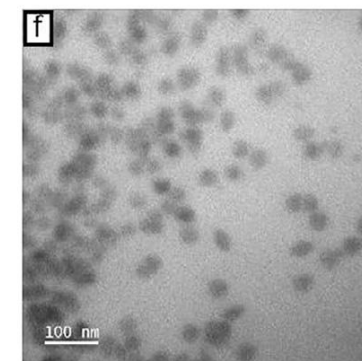
Metal-organic Frameworks, CAS # 10027-30-2

Metal-organic frameworks (MOFs) are a class of porous materials composed of metal ions or clusters coordinated to organic ligands, forming a crystalline structure. MOFs are renowned for their exceptionally high surface area, tunable pore sizes, and structural diversity, which make them ideal for various applications. They are used in gas storage and separation (e.g., CO₂ capture), catalysis, drug delivery, and as components in sensors and electronic devices.

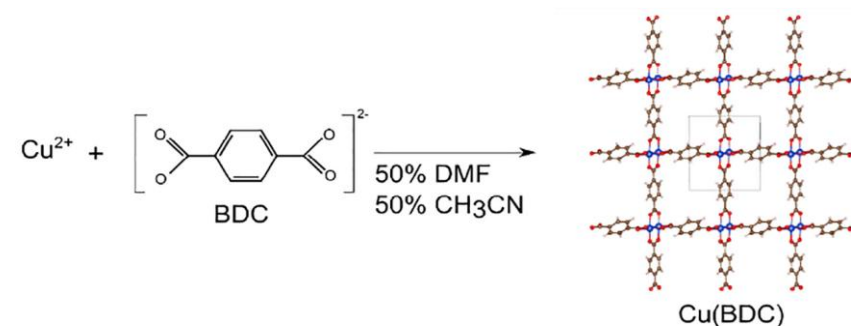
Property	Flakes
Formula	CuBDC
Crystal phase	CuBDC
Particle size, nm	32
Purity, %	98+
Application	Catalysis, adsorption
Function	Catalyst support, adsorbent
Specific surface area, m ² g ⁻¹	324



Particle size distribution, TEM



TEM of dispersed particles

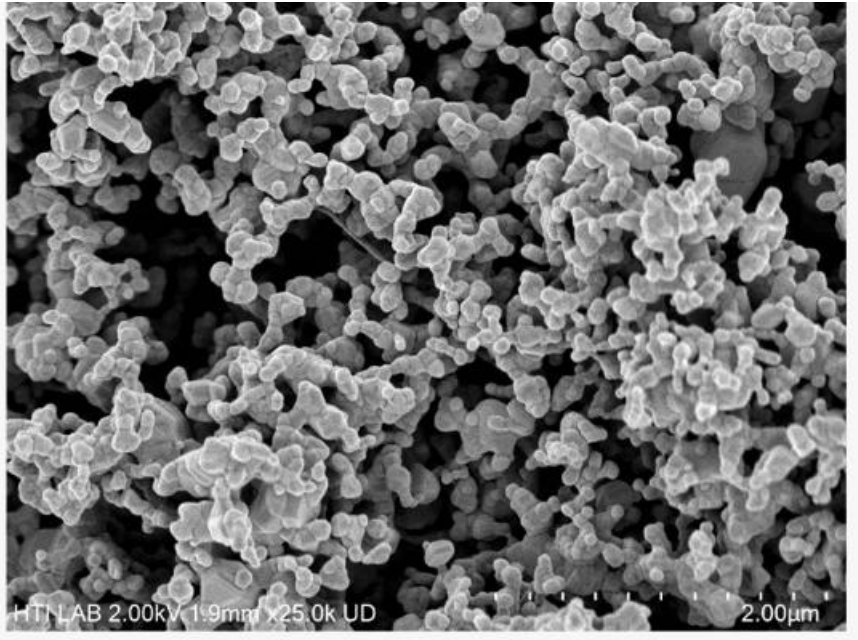


Reference: <https://doi.org/10.1016/j.cej.2020.124133>

Silver, CAS # 7440-22-4

Silver nanopowder consists of nanoparticles of silver, typically ranging in size from 1 to 100 nanometers. Due to its high surface area and unique properties at the nanoscale, silver nanopowder exhibits excellent electrical conductivity, thermal conductivity, and antimicrobial activity. These properties make it valuable in various applications, including electronics for conductive inks and pastes, in medical devices and coatings for its antimicrobial effects, and in catalysis.

Property	Spheres
Formula	Ag
Crystal phase	Cubic
Particle size, nm	5-150
Purity, %	98+
Application	Conductive inks
Function	Conductive material



HR-SEM of Silver nanopowder